One Divide, Ten Questions: The Geography of Alberta's 2026 Referendum

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1 Methodology

1.1 Data

We use the official results of the 19 October 2026 Alberta referendum, published by Elections Alberta for each of the province's 87 electoral divisions (ridings) (Elections Alberta 2026). The ballot carried ten questions (Table 1). Questions 1 to 5 concerned immigration and voting rules and were not binding. Questions 6 to 9 asked whether the provincial government should seek constitutional amendments with other provinces. Question 10 asked voters to choose between Alberta remaining a province (Option 1) and the government beginning the legal process for a binding referendum on separation (Option 2). It was not itself binding.

Table 1: The ten referendum questions. Short labels are used in the figures and tables that follow.

Short label	Wording on the ballot
Q1 Immigration control	Do you support the Government of Alberta taking increased control over immigration for the purposes of decreasing immigration to more sustainable levels, prioritizing economic migration and giving Albertans first priority on new employment opportunities?
Q2 Program eligibility	Do you support the Government of Alberta introducing a law mandating that only Canadian citizens, permanent residents and individuals with an Alberta-approved immigration status will be eligible for provincially-funded programs, such as health care, education and other social services?
Q3 Residency wait	Assuming that all Canadian citizens and permanent residents continue to qualify for social support programs as they do now, do you support the Government of Alberta introducing a law requiring all individuals with a non-permanent legal immigration status to reside in Alberta for at least 12 months before qualifying for any provincially-funded social support programs?
Q4 Health/education fee	Assuming that all Canadian citizens and permanent residents continue to qualify for public health care and education as they do now, do you support the Government of Alberta charging a reasonable fee or premium to individuals with a non-permanent immigration status living in Alberta for their and their family's use of the health care and education systems?
Q5 Citizenship to vote	Do you support the Government of Alberta introducing a law requiring individuals to provide proof of citizenship, such as a passport, birth certificate or citizenship card, to vote in an Alberta provincial election?
Q6 Provinces pick judges	Do you support the Government of Alberta working with the governments of other willing provinces to amend the Canadian Constitution to have provincial governments, and not the federal government, select the justices appointed to provincial King's Bench and Appeal courts?

Table 1: The ten referendum questions. Short labels are used in the figures and tables that follow.

Short label	Wording on the ballot
Q7 Abolish Senate	Do you support the Government of Alberta working with the governments of other willing provinces to amend the Canadian Constitution to abolish the unelected federal Senate?
Q8 Opt out with funding	Do you support the Government of Alberta working with the governments of other willing provinces to amend the Canadian Constitution to allow provinces to opt out of federal programs that intrude on provincial jurisdiction such as health care, education and social services, without a province losing any of the associated federal funding for use in its social programs?
Q9 Provincial paramountcy	Do you support the Government of Alberta working with the governments of other willing provinces to amend the Canadian Constitution to better protect provincial rights from federal interference by giving a province's laws dealing with provincial or shared areas of constitutional jurisdiction priority over federal laws when the province's laws and federal laws conflict?
Q10 Separation process	Should Alberta remain a province of Canada, or should the Government of Alberta commence the legal process required under the Canadian Constitution to hold a binding provincial referendum on whether or not Alberta should separate from Canada?

The division-level workbooks were cleaned into one row per division and question. For every question we use valid votes only, the sum of the two choices, and exclude declined and rejected ballots. The cleaned totals match the provincial figures in Elections Alberta's 29 October 2026 news release exactly for Question 10, and to within about 100 votes on every other question (the largest gap is on Question 7, where the workbooks and the news release disagree).

Division boundaries are the 2017 electoral divisions, which were in force for the 2026 vote, from the Government of Alberta's geospatial service (Government of Alberta 2017). The boundaries were simplified on download (to a tolerance of 0.002 degrees) and repaired where simplification left invalid polygons. Results and boundaries are joined by division number, not name, because one division is named Calgary-McCall in the boundary file and Calgary-Bhullar-McCall in the results.

Each riding is assigned to one of five regions (Figure 1). Calgary and Edmonton contain only the ridings within those cities; suburbs and nearby towns such as St. Albert, Sherwood Park, Airdrie and Chestermere are part of Central. These labels were carried over from an earlier exploratory notebook and have not been checked against an official classification.

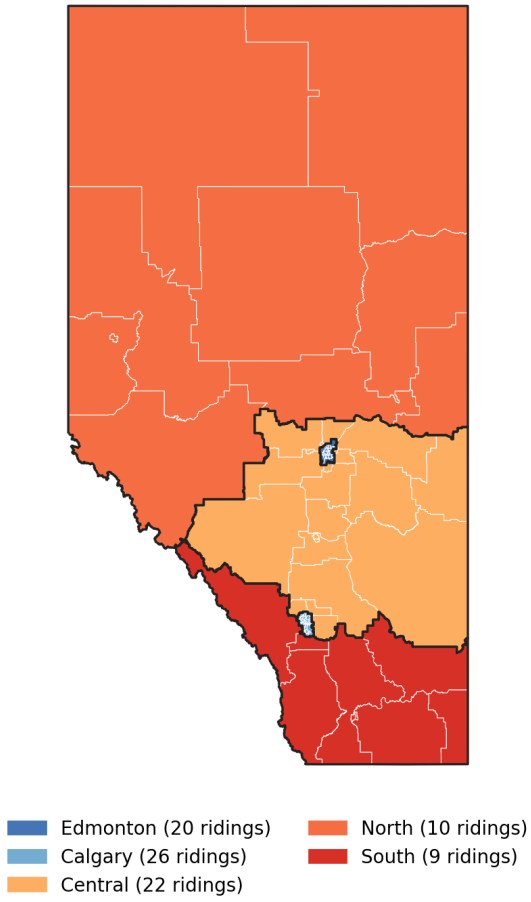


Figure 1: The five regions used in the analysis. Thin white lines are riding boundaries; dark lines are region boundaries. Calgary (26 ridings) and Edmonton (20) are the small areas inside Central and South.

To compare the referendum with party support, we use Elections Alberta's official results for the 2019 and 2023 provincial general elections (Elections Alberta 2023), which were held on the same 2017 boundaries, so every riding matches exactly. A party's share is its votes as a percentage of valid ballots in the riding. In each riding the candidate votes sum to the valid ballots, and the provincial totals reproduce the published results (UCP 52.6% and NDP 44.0% in 2023; UCP 54.9% and NDP 32.7% in 2019). Three small parties serve as indicators of separatist sentiment: the Alberta Independence Party in 2019, which ran in 63 ridings, and The Independence Party of Alberta and the Wildrose Independence Party in 2023, which together ran in 15. Where a party had no candidate, the riding is excluded from that comparison rather than counted as zero.

Turnout is ballots cast divided by the elector count printed in the division workbooks. Those elector counts are higher than the count in the news release, so riding-level turnouts are a few points below the headline provincial figure of 70.4%.

1.2 Measures

For Question 10, the outcome is the **Option 2 share**: Option 2 votes as a percentage of valid votes in a riding. Because the two options sum to 100%, every result for Option 2 applies in mirror image to Option 1, and Moran's I is identical for both.

To compare questions we define **support** as the Yes share for questions 1 to 9 and the Option 2 share for Question 10. On every question this is the side asking for more provincial authority or for change, so support

can be compared directly across the ballot.

1.3 Spatial dependence

Neighbours are defined by **queen contiguity**: two ridings are neighbours if their boundaries touch at an edge or a corner. To absorb small gaps left by boundary simplification, ridings within 500 m of each other in NAD83 / Alberta 10-TM (a metric projection) are treated as touching. Every riding has at least one neighbour. The weights matrix W is row-standardised. As a sensitivity check we also use the 3, 5, 7 and 10 nearest ridings, measured between centroids in the same projection.

Global clustering is measured with Moran's I (Moran 1950),

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2},$$

where y_i is the outcome in riding i. Significance comes from a permutation test: the observed values are reassigned to ridings at random 9,999 times, and the one-sided p-value is the share of permutations with an I at least as large as observed (so the smallest attainable value is 0.0001).

Ridings in the same region are both similar and adjacent, so a high Moran's I could reflect regional differences alone. To separate the two, we subtract each region's mean from its ridings and compute Moran's I on the residuals, which measures clustering *within* regions.

Local clustering uses local Moran's I (Anselin 1995). Each riding is classified by its own standardised value and the average of its neighbours' as High-High (a hotspot), Low-Low (a coldspot), High-Low or Low-High. Significance uses conditional permutation: the riding's own value is held fixed and its neighbours are replaced by random draws from the other ridings, 9,999 times. Ridings with a pseudo p-value of 0.05 or more are reported as not significant. With 87 tests at this level about four false positives are expected by chance, so we interpret the overall pattern rather than individual borderline ridings.

1.4 Comparing the ten questions

Agreement between questions is measured by the Pearson correlation of riding-level support across the 87 ridings, and by the share of total variance captured by the first principal component of the ten-by-ten correlation matrix. Region-level support is computed from summed votes, so larger ridings carry more weight.

To find ridings where Question 10 departed from the rest of the ballot, we regress each riding's Question 10 support on its average support across questions 1 to 9, using a quadratic fit because the relationship curves at low levels of support. The residual is the riding's **departure**: the number of percentage points by which its Question 10 result was above or below what its other answers predict. We test whether departures are themselves clustered with Moran's I. We also compute a **carry-over ratio**, Question 10 support divided by average support on questions 1 to 9, which expresses how much of a riding's support for the other proposals extended to separation.

1.5 The referendum and the party vote

For each question we report the correlation across ridings between support and the UCP's 2023 share, and the difference between them: how far a question's support ran above or below the governing party's own vote in the same place. Region-level differences use summed votes for both the referendum and the election. To test whether the party vote accounts for the referendum's geography, we fit Question 10 support on the 2023 UCP share with a quadratic and compute Moran's I on the residuals, as for the departures above.

To test whether past votes mark the ridings that departed from the pattern on Question 10, we correlate the departures with five election measures: the Alberta Independence Party share in 2019, the combined share of the two 2023 independence parties, the Freedom Conservative Party share in 2019, the change in the UCP share from 2019 to 2023, and the UCP share in 2023. Significance comes from 9,999 random permutations

of one variable, two-sided. We report all five tests and make no correction for multiple comparisons, so individual p-values near 0.05 should be read with caution.

Elections and referendums have different electorates and turnout, three years apart. A riding's referendum support falling below its UCP share does not show that UCP voters in that riding voted against a proposal; it shows that the proposal did worse there than the party had.

1.6 Turnout

The association between turnout and Option 2 share is measured with Spearman's rank correlation, across all ridings and within each region.

1.7 Scope and software

All quantities are measured at the riding level. They describe places, not voters: a riding that supported two questions did not necessarily do so through the same people, and no claim here should be read as a statement about individual behaviour. The analysis uses Python (Van Rossum and Drake 2009) with pandas (McKinney 2010), GeoPandas (Jordahl et al. 2020) and Matplotlib (Hunter 2007). Every number, table and figure in this paper is produced by `notebooks/06-spatial_analysis.ipynb`.

2 Results

2.1 Separation lost almost everywhere, though the map makes it look closer

Option 2 received 520,233 votes to Option 1's 1,556,316, or 25.1% of valid votes. It won a majority in 2 of 87 ridings, Rimbey-Rocky Mountain House-Sundre (50.9%) and Central Peace-Notley (50.5%), both by less than one percentage point. Its weakest result was 9.3% in Edmonton-Strathcona.

The divide was between the two large cities and everywhere else (Figure 2). No riding in Calgary or Edmonton gave Option 2 more than 23.3%, below the provincial result, while the median North riding gave it 41.0% and the median Central riding 32.0%. Every one of the 20 strongest ridings for Option 2 lies outside the two cities.

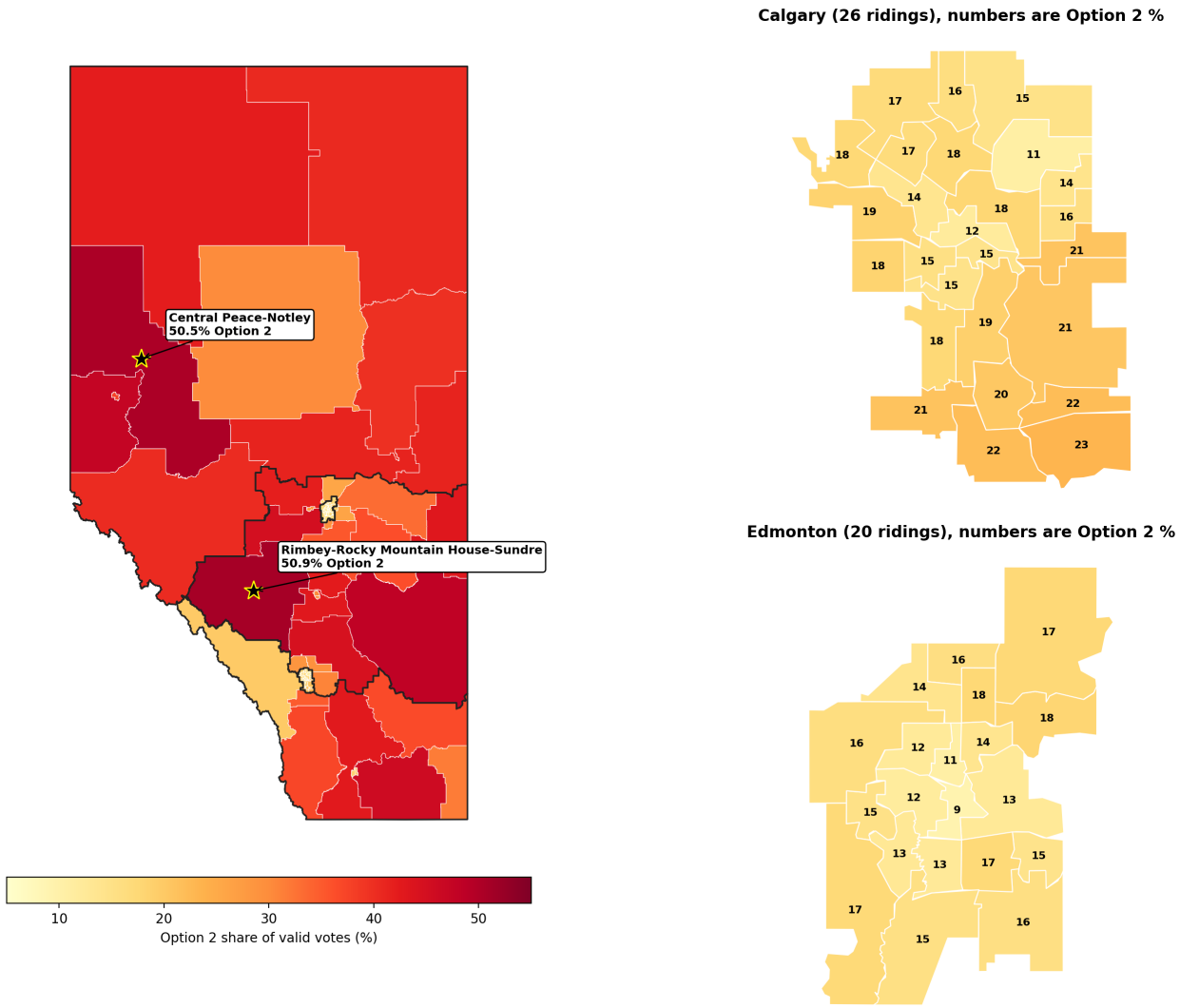


Figure 2: Option 2 share of valid votes on Question 10, by riding, with Calgary and Edmonton enlarged (numbers are Option 2 percentages). Stars mark the two ridings where Option 2 won a majority.

A map coloured by riding overstates Option 2, because the ridings that supported it most cover most of the province's land (Figure 3). The North covers 68% of Alberta's land area but cast 8.5% of valid votes. Calgary and Edmonton together cover well under 1% of the land but cast 52% of valid votes, and even with low Option 2 shares they supplied 34% of all Option 2 votes. The largest single source of Option 2 votes was Central Alberta, with 39% of the total from 29% of the vote. The largest block of the separation vote, in other words, came from Central Alberta, the rural areas, towns and suburbs between and around the two cities, rather than from the sparsely populated north.

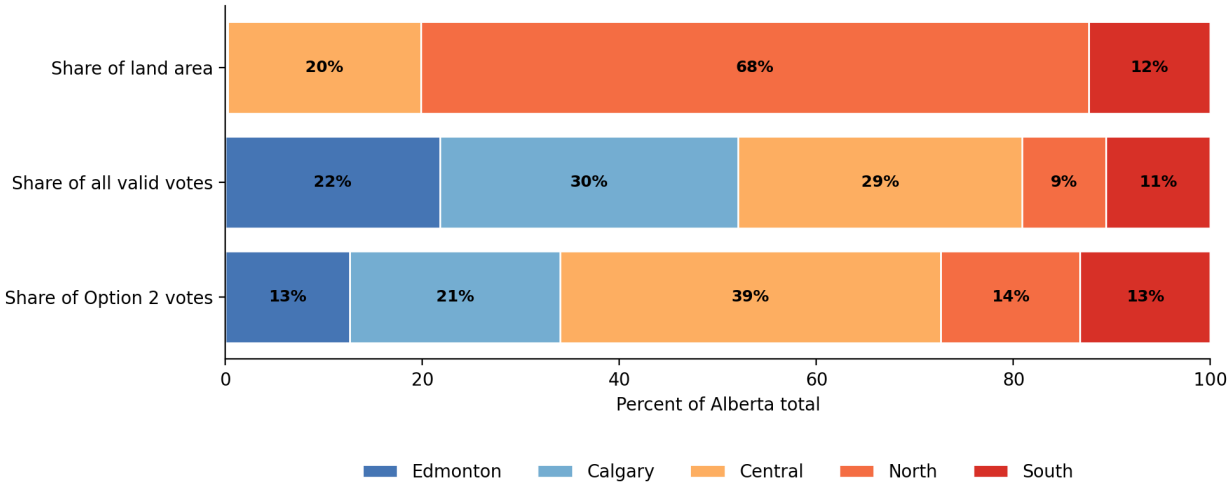


Figure 3: Each region's share of Alberta's land, of all valid votes on Question 10, and of Option 2 votes.

2.2 Support is clustered, and the clusters are regions

Neighbouring ridings voted alike. With shared-border neighbours, Moran's I for the Option 2 share is 0.73. None of the 9,999 random reassignments came close (the largest was 0.26), giving $p = 0.0001$, the smallest attainable value. The result does not depend on how neighbours are defined: with the 3 to 10 nearest ridings, I ranges from 0.49 to 0.67, falling as more distant ridings are counted as neighbours.

Most of this clustering is the difference between regions. After subtracting regional means, Moran's I falls to 0.11 ($p = 0.033$). Some clustering remains within regions, but it is weak: a riding's region predicts most of what its neighbours' results would.

The local analysis shows the shape of the divide (Figure 4). 22 ridings are significant hotspots: every riding in the North (10), 9 in rural Central Alberta and 3 in the southeast around Medicine Hat, Brooks and Cardston. They form one contiguous area. 20 ridings are significant coldspots, all in the older, central parts of Edmonton (12) and Calgary (8). No riding is a significant outlier. The remaining 45 ridings, including the outer city ridings, the suburbs and towns around both cities, Red Deer, Lethbridge and the southwest foothills, sit between the two blocs.

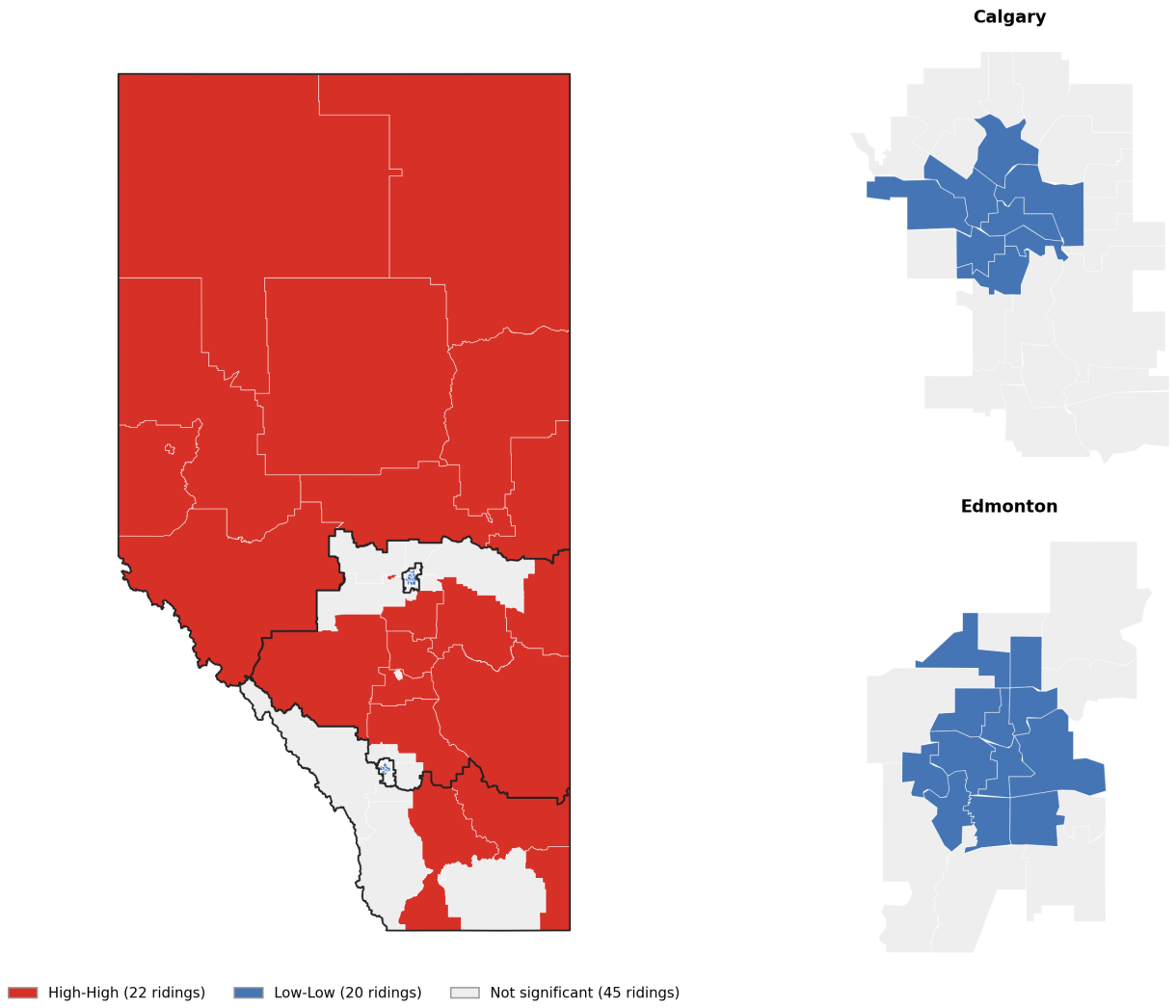


Figure 4: Significant local clusters of Option 2 support (conditional permutation, $p < 0.05$), with Calgary and Edmonton enlarged. Grey ridings show no pattern beyond chance.

2.3 Ten questions, one map

The other nine questions split the province in the same way. Questions 1 to 5 won majorities province-wide and questions 6 to 9 did not (Table 2). Support ranged from 58.1% on Q5 Citizenship to vote to 25.1% on Q10 Separation process, which was 17 points below any other question.

Table 2: Province-wide support, riding-level range and spatial clustering for each question. Support is Yes for questions 1 to 9 and Option 2 for Question 10. Moran's I uses shared-border neighbours; every value has p=0.0001. 'Within regions' is Moran's I after removing regional means.

Question	Support (%)	Ridings > 50%	Lowest (%)	Highest (%)	Moran's I	I within regions
Q1 Immigration control	54.9	48	27.5	78.5	0.72	0.17
Q2 Program eligibility	52.5	42	24.7	76.8	0.72	0.18
Q3 Residency wait	53.1	43	25.7	77.3	0.73	0.19
Q4 Health/education fee	51.8	41	24.5	76.3	0.70	0.15

Table 2: Province-wide support, riding-level range and spatial clustering for each question. Support is Yes for questions 1 to 9 and Option 2 for Question 10. Moran's I uses shared-border neighbours; every value has p=0.0001. 'Within regions' is Moran's I after removing regional means.

Question	Support (%)	Ridings > 50%	Lowest (%)	Highest (%)	Moran's I	I within regions
Q5 Citizenship to vote	58.1	62	30.5	81.5	0.72	0.17
Q6 Provinces pick judges	45.4	33	19.5	74.5	0.72	0.15
Q7 Abolish Senate	47.3	34	22.0	75.0	0.72	0.15
Q8 Opt out with funding	42.4	27	17.2	69.3	0.73	0.15
Q9 Provincial paramountcy	44.2	31	18.5	72.5	0.74	0.15
Q10 Separation process	25.1	2	9.3	50.9	0.73	0.11

Within every question, ridings disagreed widely, by 42 to 55 percentage points between the lowest and highest riding (Figure 5). In every case the city ridings sit at the low end and the North and Central ridings at the high end. What changes from question to question is where 50% falls in that ordering: questions 1 to 5 won a majority in 41 to 62 ridings, questions 6 to 9 in 27 to 34, and Question 10 in 2.

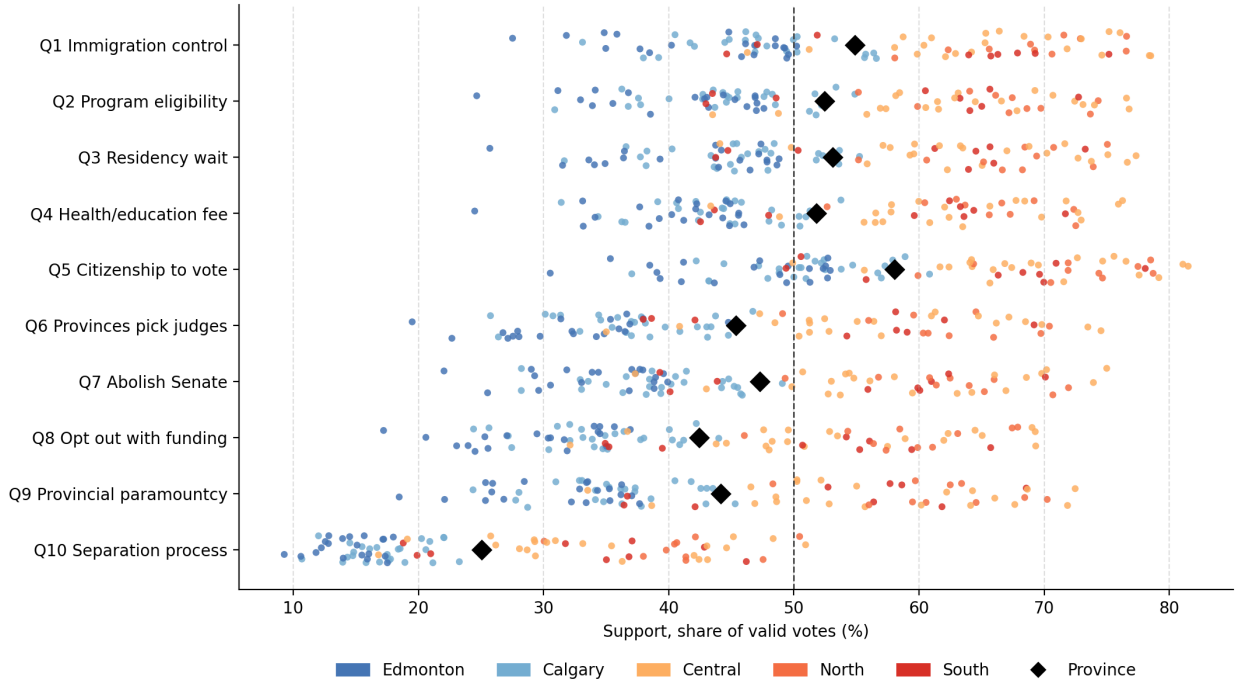


Figure 5: Support on every question in every riding, coloured by region. Diamonds are the province-wide results; the dashed line is 50%.

The questions are close to interchangeable as maps (Figure 6). Every pair of questions correlates at 0.96 or higher across ridings, and a single shared pattern accounts for 99% of the variation in support across all ten questions. Every question is clustered to the same degree (Moran's I from 0.70 to 0.74), and on every question most of that clustering is regional (0.11 to 0.19 within regions). The regions keep the same order on all ten questions (Figure 7): the North gave the most support and Edmonton the least, the gap between them was 26 to 31 points, and Calgary was 3 to 5 points more supportive than Edmonton throughout.

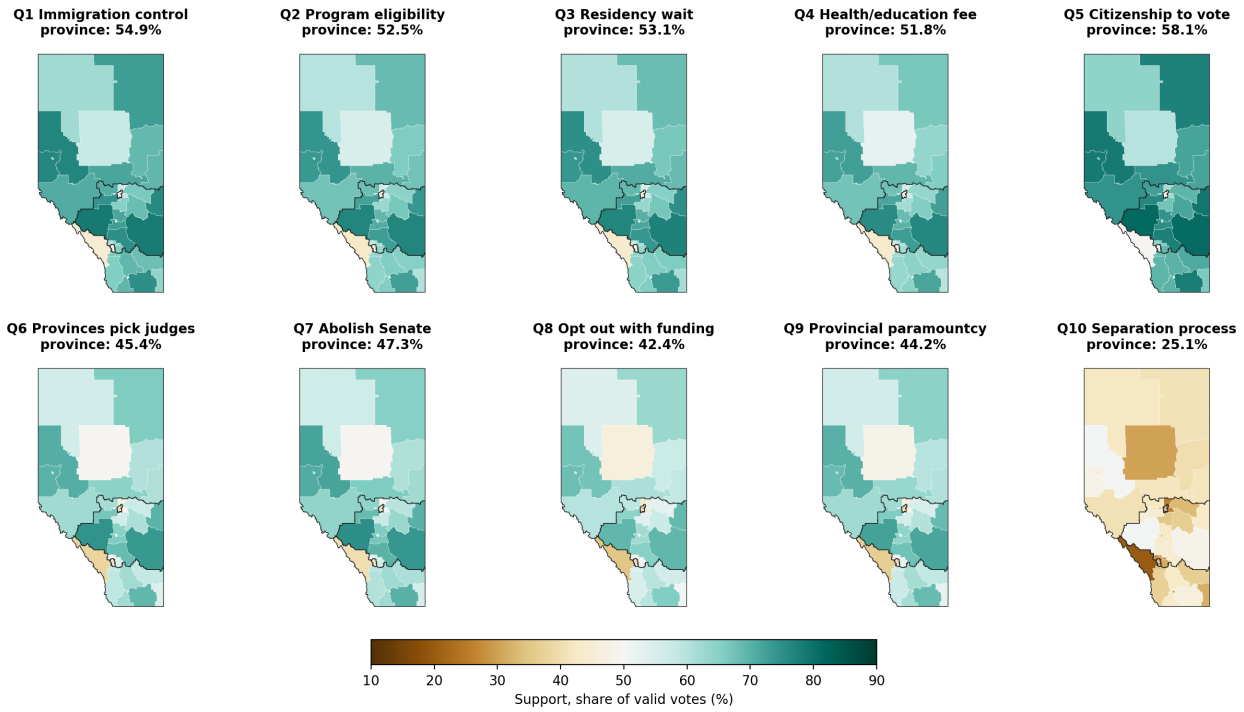


Figure 6: Support for each question by riding, on a common scale. Green is majority support and brown majority opposition.

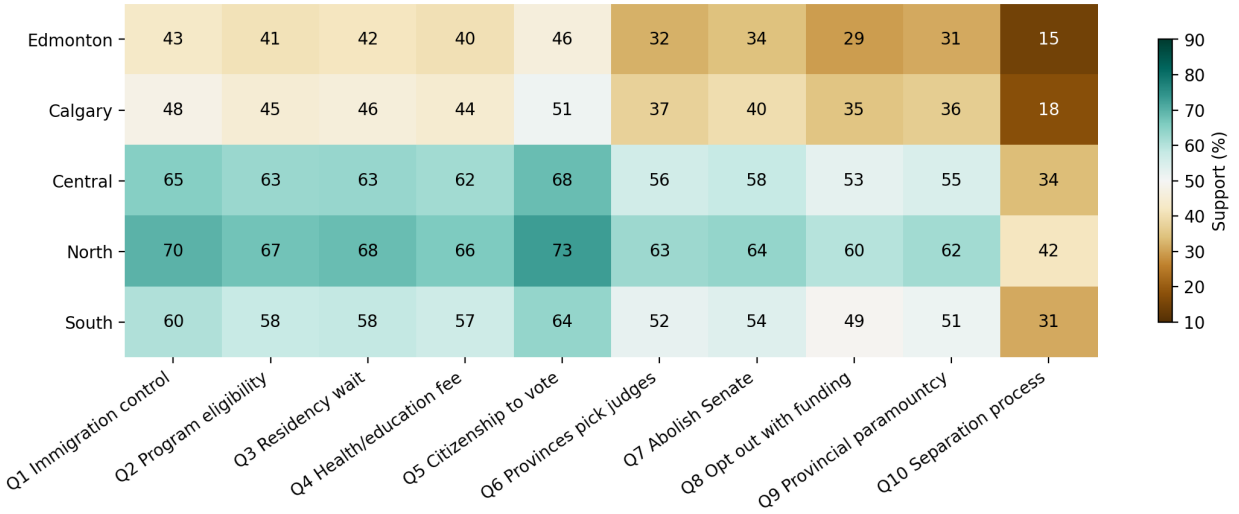


Figure 7: Support by region and question, from summed votes.

The ballot therefore did not reveal ten separate geographies of opinion. It measured one urban–rural divide ten times, at different levels. The two groups of questions differ slightly in where their support came from: the gap between the immigration and voting questions and the constitutional questions was widest in Edmonton (10.7 points) and Calgary (9.7) and narrowest in the North (6.9). City voters were relatively more open to the immigration proposals than to constitutional change.

2.4 The referendum followed the governing party's vote, at different heights

The single geography of the referendum is close to the geography of the UCP vote. Across ridings, every question's support correlates with the party's 2023 share at 0.93 to 0.95 (Figure 8). The questions differ in where they sit against that vote. The immigration and voting questions ran close to it: their support exceeded the UCP share in 35 to 77 of 87 ridings. The constitutional questions ran below it, exceeding the UCP share in at most 12 ridings and falling 5 to 10 points short on average. Question 10 fell below the UCP share in all 87 ridings, by 27 points on average.

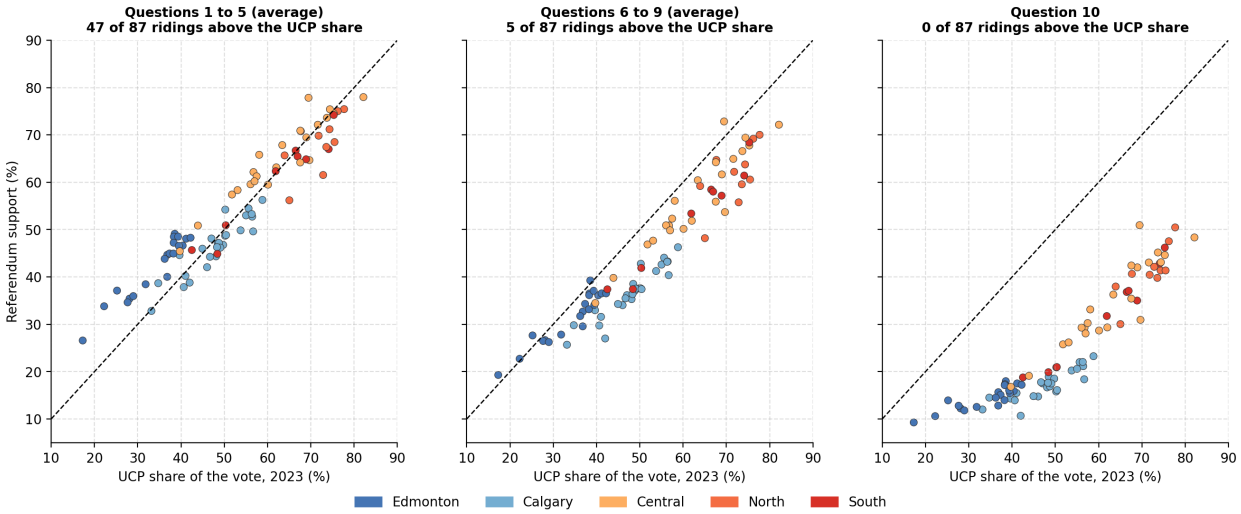


Figure 8: Referendum support against the UCP's 2023 share, by riding. The dashed line marks equal values; points above it are ridings where the question did better than the UCP had.

The regional breakdown shows where each group of questions found support beyond the party (Figure 9). In Edmonton, where the UCP took 34% of the 2023 vote, the immigration and voting questions ran 6 to 12 points above it: these proposals drew support from well outside the governing party's electorate. Elsewhere they ran within a few points of the UCP vote. The constitutional questions fell short of the UCP vote in every region. Question 10 fell short by 20 to 31 points, most in Calgary and the North. In the North, where the UCP took 72% of the vote, separation drew 42%.

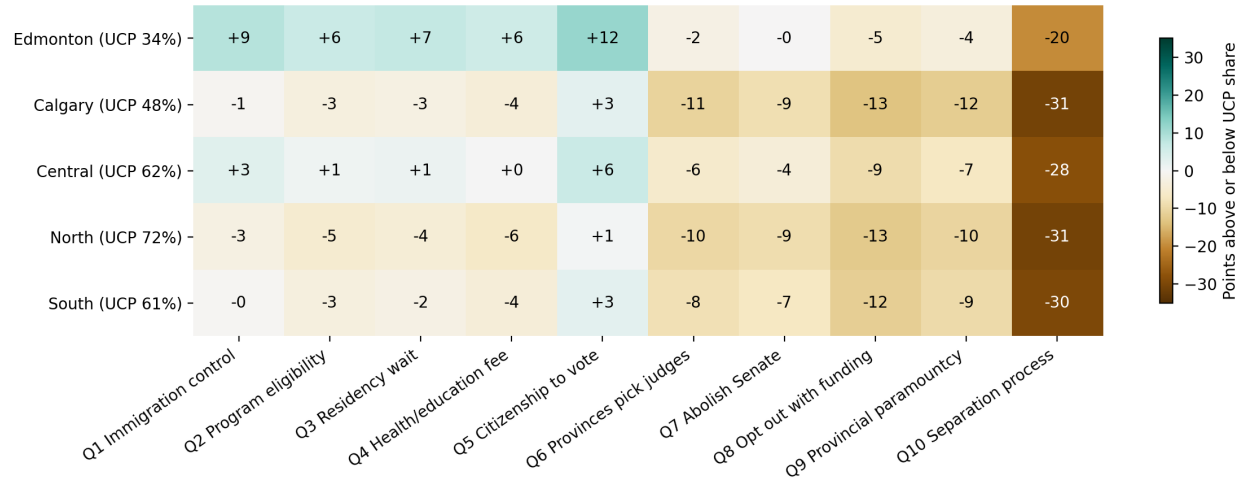


Figure 9: Referendum support minus the UCP's 2023 share, by region and question, from summed votes. Regions are labelled with the UCP's 2023 share.

The party vote does not, however, account for all of the referendum's geography. The 2023 UCP share explains 92% of the variation in Question 10 support across ridings, but the remainder is still spatially clustered (Moran's I = 0.36, p = 0.0001). Neighbouring ridings miss the party-based prediction in the same direction, so some of the regional pattern in the separation vote is not a reflection of 2023 partisanship. The other referendum questions predict Question 10 better (97%, with leftover clustering of 0.19).

2.5 Separation sits at the far end of the same divide, below a ceiling

Question 10 drew from the same places as every other question, but its support was not a fixed fraction of theirs. In the 27 ridings where all nine other proposals won a majority, Option 2 ranged from 30.9% to 50.9%, with a median of 41.4%; only 2 crossed 50%. In the 24 ridings where none of the nine won a majority, almost all in the two cities, the median was 14.2%.

The carry-over ratio shows how this ceiling varied (Figure 10). The median Edmonton riding gave Question 10 0.38 times its average support for the other proposals, and the median Calgary riding 0.42, against 0.53 in Central, 0.57 in the South and 0.62 in the North. Across ridings, carry-over rose almost in step with support for the other proposals (correlation 0.92). Where the other proposals were popular, Question 10 kept more than half of their level of support; where they were unpopular, it kept well under half. Even in its strongest territory, separation fell well short of the majorities the other proposals won there.

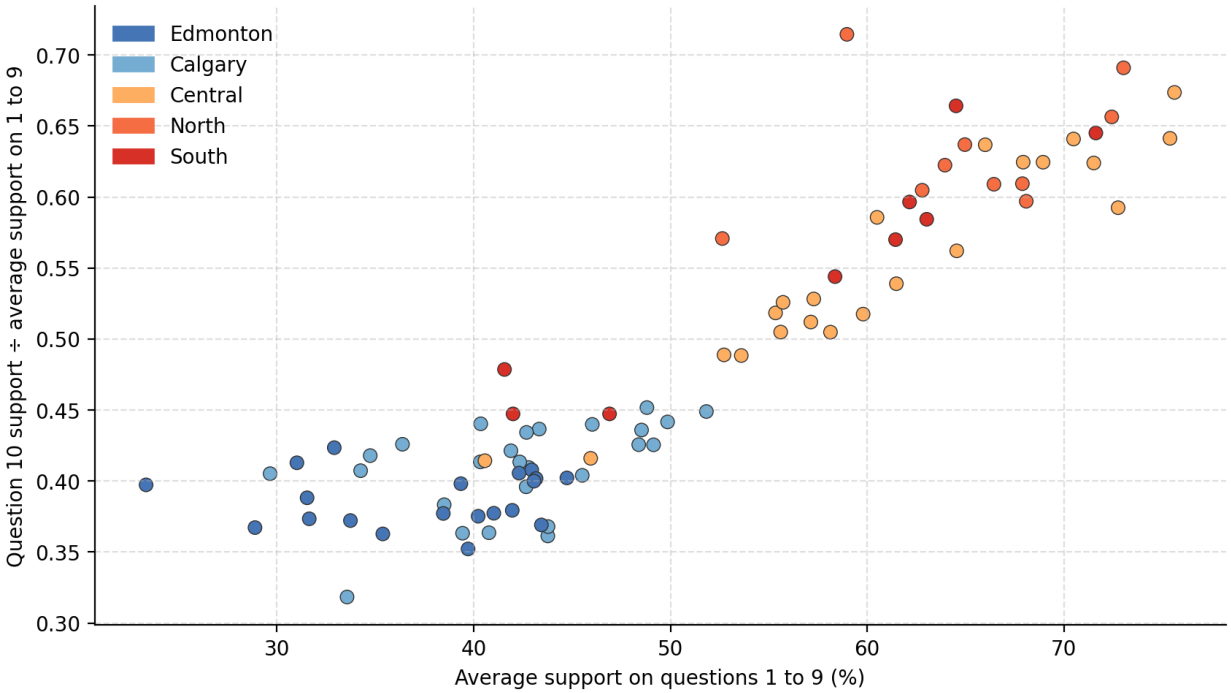


Figure 10: Question 10 support as a fraction of each riding's average support on questions 1 to 9, against that average.

2.6 Where separation broke from the pattern

Because Question 10 tracked the other questions so closely, the ridings that departed from that relationship are informative. A riding's average support on questions 1 to 9 accounts for 97% of the variation in Question 10 support (Figure 11). The departures are small, mostly within a few points, but they are not random: they cluster geographically (Moran's I = 0.19, p = 0.0016).

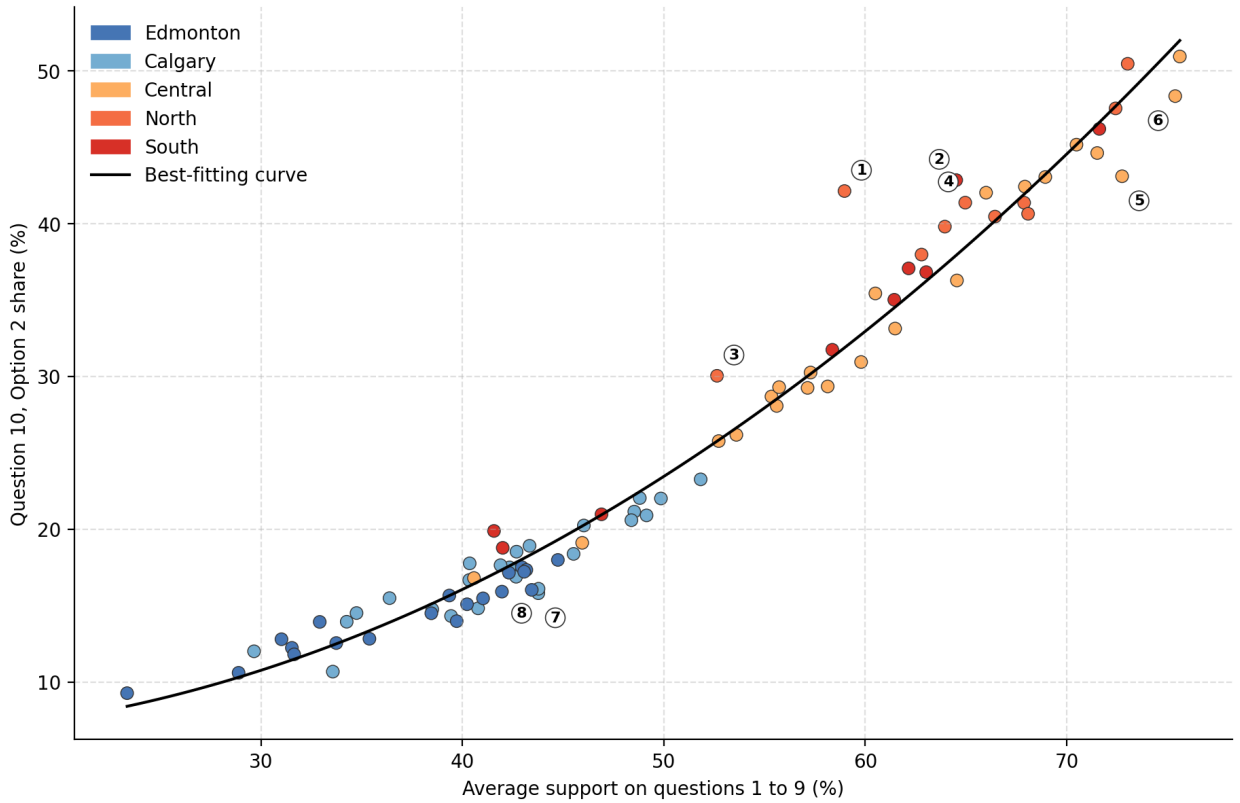


Figure 11: Question 10 support against average support on questions 1 to 9, with a quadratic fit. Numbered ridings are the four furthest above and below the curve, listed in Table 3.

Table 3: Ridings whose Question 10 result departed most from what their answers to questions 1 to 9 predict. Positive values mean more Option 2 support than predicted.

Label	Riding	Region	Departure (points)
1	Peace River	North	+10.3
2	Cardston-Siksika	South	+4.9
3	Lesser Slave Lake	North	+4.3
4	Bonnyville-Cold Lake-St. Paul	North	+2.9
5	Vermilion-Lloydminster-Wainwright	Central	-5.0
6	Drumheller-Stettler	Central	-3.3
7	Calgary-Cross	Calgary	-2.8
8	Calgary-North	Calgary	-2.5

The largest positive departure was Peace River, where Option 2 ran 10.3 points above prediction, followed by Cardston-Siksika (South, +4.9), Lesser Slave Lake (North, +4.3), Bonnyville-Cold Lake-St. Paul (North, +2.9). On average, northern ridings gave separation 2.2 points more than their other answers implied, and southern ridings 1.3 points more (Figure 12). The largest negative departures were in east-central Alberta, led by Vermilion-Lloydminster-Wainwright (-5.0 points), and in a group of Calgary ridings. These ridings stopped further short of separation than their answers on the other nine questions would predict.

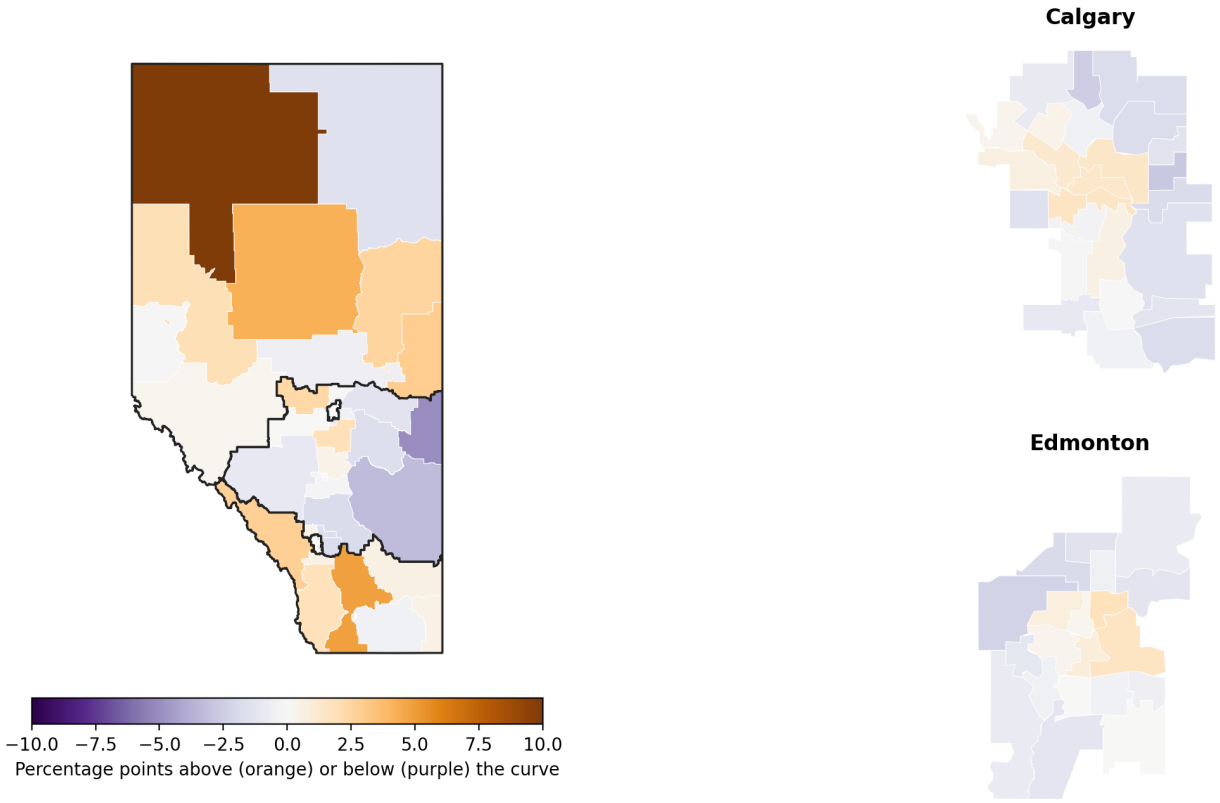


Figure 12: Departure of each riding's Question 10 result from the fitted curve, with Calgary and Edmonton enlarged. Orange ridings gave Option 2 more support than their other answers predict; purple ridings gave less.

Past elections only faintly mark these ridings (Table 4). Departures were larger where the Alberta Independence Party did better in 2019 (correlation 0.27 across 63 ridings, p = 0.033), although that party never won more than 2.5% of a riding's vote (Figure 13). They were also larger where the UCP's share rose more, or fell less, between 2019 and 2023 (correlation 0.25, p = 0.018), the first election after Danielle Smith became party leader. The UCP's 2023 share itself was unrelated to the departures, so the association is with the movement of the party's vote, not its level. Both correlations are weak, each accounting for less than a tenth of the variation, and they are two significant results among five uncorrected tests. They suggest where to look rather than explain the departures; doing so would require data on who lives in each riding.

Table 4: Correlation between each riding's Question 10 departure and election measures, with two-sided permutation p-values. Ridings without a candidate from the party are excluded.

Measure	Ridings	Correlation with departure	p
Alberta Independence Party share, 2019	63	+0.27	0.033
Independence parties' share, 2023	15	-0.16	0.502
Freedom Conservative Party share, 2019	24	-0.20	0.224
Change in UCP share, 2019 to 2023	87	+0.25	0.018
UCP share, 2023	87	+0.09	0.392

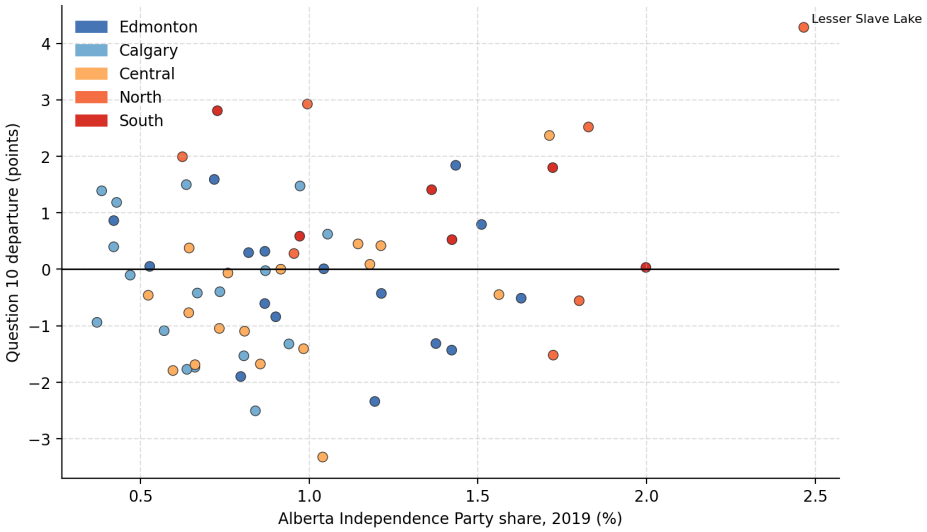


Figure 13: Question 10 departure against the Alberta Independence Party's 2019 share, in the 63 ridings where it ran.

2.7 Turnout does not account for the pattern

Turnout varied from 50.8% in Edmonton-Decore to 80.7% in Sherwood Park, and was lowest in the North (median 58.0%). Across all ridings, the rank correlation between turnout and Option 2 share was -0.18. Within regions the relationship was sometimes moderate but ran in opposite directions, positive in the North (0.61) and Calgary (0.31) and negative in the South (-0.70), Central (-0.62) and Edmonton (-0.32), with 9 to 26 ridings per region. There is no consistent association between how many people voted in a riding and how it voted on separation.

Anselin, Luc. 1995. "Local Indicators of Spatial Association—LISA." *Geographical Analysis* 27 (2): 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x.

Elections Alberta. 2023. *Official Results of the 2019 and 2023 Provincial General Elections by Electoral Division*. Https://officialresults.elections.ab.ca.

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